

Kourosh Sajjadi

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Education

University of Tehran

B.Sc. in Computer Engineering (Software Engineering Concentration),
School of Electrical and Computer Engineering, College of Engineering

Tehran, Iran

Sep. 2021 – Sep. 2026

- GPA: **18.93/20**; ranked among the **top 5%** of B.Sc. students in Computer Engineering.

Selected coursework

- Neural Networks and Deep Learning (20/20)
- Fundamentals of Computer Vision (20/20)
- Research and Technical Presentation (20/20)
- Artificial Intelligence (20/20)
- Fundamentals of Data Science (20/20)
- Data Structures and Algorithms (20/20)

Research interests

- AI Evaluation & Reliability
- AI for Software Engineering
- Reliable AI Systems
- Multimodal Sensing & Perception
- Computer Vision
- AI in Education & Intelligent Tutoring

Research experience

Research Assistant (B.Sc. Thesis)

Incremental Code Quality Review Using Large Language Models

Oct. 2025 – Present

University of Tehran

- Expanded an undergraduate software-engineering project into my B.Sc. thesis, developing a Git-based workflow for incremental code-quality review using large language models to support human evaluators rather than replace them.
- Combined project requirements, evaluation criteria, domain documentation, and code changes with evidence validation and traceable reporting; evaluated the workflow on real Java and Go projects and compared different ways of providing project-specific context.

Research Assistant

Post-Earthquake Building Damage Assessment for Disaster Response

Oct. 2025 – Present

University of Tehran

- Studied building-level damage assessment from pre- and post-earthquake remote-sensing imagery to support faster identification and prioritization of affected structures for disaster response.
- Designed and evaluated an approach that first identifies affected buildings and then estimates damage severity; across four earthquake events and 373 test tiles, it improved affected-building precision and recall to 61.07% and 33.81%, compared with 53.66% and 25.98% for the reference baseline on common support.

Research Assistant

Simulated Persian Students for Evaluating AI Tutoring Systems

Feb. 2026 – Present

University of Tehran

- Designed LLM-simulated Persian student personas with bounded knowledge, stable misconceptions, confidence, engagement, and help-seeking behavior to test AI tutoring systems before enough real student dialogue is available.
- Built a 5–10-turn LMS evaluation workflow that records conversations and session metadata to assess persona consistency, misconception stability, and tutor responses across student profiles; the resulting dialogue data supports controlled evaluation and future personalization.

Languages

English: Full professional proficiency; IELTS Academic 8.0
(CEFR C1; Listening 9.0, Reading 8.5, Speaking 8.0, Writing 7.0), Oct. 2025

Persian: Native

Honors and awards

- Best Undergraduate Project Award:** University of Tehran, School of Electrical and Computer Engineering, Project Day 2026; awarded for the project later developed into my B.Sc. thesis.
- Benefactors Foundation Scholarship:** 8th University of Tehran scholarship for talented students, 2022–2023.
- Faculty of Engineering Award:** University of Tehran; recipient in the 2021–2022 academic year and nominated again for the award in the following cycle.
- National entrance examination:** Ranked top 0.5% in the Iranian Nationwide University Entrance Examination.

Teaching experience

Teaching Assistant (programming assignments, tutorials, and homework/exam grading) <i>School of Electrical and Computer Engineering, University of Tehran</i>	2023 – 2026 <i>Tehran, Iran</i>
Artificial Intelligence	Spring 2026, Fall 2025, Spring 2025
Data Structures and Algorithms	Fall 2024, Spring 2024, Fall 2023
Principles of Database Design	Fall 2025, Spring 2025
Compiler Design and Programming Languages	Fall 2025, Fall 2024
Computer Networks	Spring 2025, Fall 2025

Professional experience

Full-Stack Developer and AI Engineer <i>RastAI, University of Tehran Science and Technology Park</i>	Nov. 2025 – Present <i>Tehran, Iran</i>
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- Develop Kotlin and Spring backend services, Next.js and React interfaces, and Python AI components for an LLM-powered personalized learning platform.
- Build AI-assisted learning workflows covering API integration, tracing, evaluation, and observability; collaborate across the product team to turn learning requirements into implementable system designs.

Computer Vision Intern <i>University of Tehran  Repository</i>	Summer 2024 <i>Tehran, Iran</i>
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- Built Python computer-vision applications covering image processing, OCR, classification, object detection, pose estimation, and segmentation using OpenCV, MediaPipe, EasyOCR, YOLOv8, scikit-learn, Segment Anything, Streamlit, Stable Diffusion, and DeepFace.

Selected projects

Deep Learning Paper Reproductions | *Python, PyTorch, TensorFlow* | Repository

- LoRA fine-tuned the 8B-parameter LLaDA masked-diffusion model for Text-to-SQL and implemented its iterative sampling loop from scratch; reached 17.3% normalized exact match on 300 held-out queries. 
- Reproduced INC-VGGN for plant-disease classification at 92.24% accuracy versus 92.00% reported; reproduced M-Net for brain-MRI tissue segmentation and cleared all three Dice targets; implemented a VampPrior VAE study of posterior collapse.  M-Net |  VAE

Distributed Systems Course Projects | *Go, RPC, MapReduce, Raft* | Repository

- Implemented RPC services, MapReduce, Raft consensus, and sharded key-value stores in Go with tests and documentation based on MIT 6.5840/6.824-style labs.

Bird's-Eye-View Stitching System | *C++, OpenCV, Qt, Raspberry Pi* | Repository

- Built a real-time surround-view simulator with interactive calibration, an image-stitching path measured at approximately 23 ms per frame, and support for embedded deployment.

Spotify Song Recommendation System | *Python, SQLite, MLflow, Power BI* | Repository

- Built a content-based KNN recommendation pipeline spanning exploratory analysis, feature engineering, reproducible data processing, experiment tracking, persisted model artifacts, and dashboarding.

Software Engineering Matching Engine | *Java, Spring, Artemis JMS, JUnit* | Repository

- Extended a trading matching engine across several iterations with asynchronous messaging, new order types, automated tests, and design-pattern-based restructuring.

Technical skills

Programming: Python, C, C++, Java, Kotlin, Go, JavaScript, TypeScript, SQL, MATLAB, Verilog, C#

Data and infrastructure: Docker, Git, Linux, Kafka, MLflow, Langfuse, Jupyter, LaTeX, JUnit, pytest

Web and backend: Spring Boot, Next.js, React, FastAPI, REST APIs, PostgreSQL, MongoDB, SQLite, Redis

AI and machine learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8, LangChain, Haystack, OpenAI Agents SDK